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# Island Living Indirectly Affects Hair Glucocorticoid Levels in a Small Mammal

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## ABSTRACT

Island vertebrates that are small on the mainland tend to be larger and exhibit tamer behavior than their mainland conspecifics—a combined set of characteristics known as “island syndrome.” Such island-specific traits are often attributed to lower predation pressure on islands than on the mainland. While the morphology and behavior of island vertebrates has received significant attention, relatively few studies have compared physiological traits between island and mainland populations. Given that hormones such as glucocorticoids are involved in responding to predation, it is reasonable to test whether island populations that have demonstrated characteristics of island syndrome also have different glucocorticoid levels than their mainland conspecifics. Here, we used a long-term museum collection of deer mice (*Peromyscus maniculatus*) obtained from two archipelagos and two mainland regions to test the hypothesis that island syndrome includes changes in time-averaged corticosterone levels, as measured in hair. As predicted by island syndrome, deer mice from islands were structurally larger and heavier for their given structural size than their mainland conspecifics. When we compared size-matched individuals (i.e., holding size constant), hair corticosterone levels did not differ between island and mainland mice. However, corticosterone levels scaled positively with body mass and condition across our sample

population. This led to a relative increase in corticosterone levels among median-sized mice from islands relative to median-sized mice from mainland populations. We conclude that island syndrome does include effects on deer mouse stress physiology but only indirectly through the evolution of larger body size in island populations.

**Keywords:** island syndrome, stress physiology, body size, *Peromyscus*.

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## Introduction

Island vertebrates often exhibit suites of physiological, behavioral, and morphological traits that differ from those in mainland communities (Adler and Levins 1994). The combination of biased colonization, low predator and competitor abundance, and other ecological factors on islands typically results in small terrestrial vertebrates evolving toward larger body size (Benítez-López et al. 2021) and tame behavior (“island tameness”; Vitousek et al. 2010; Cooper et al. 2014). These macroevolutionary patterns have been variously described as the “island rule” or “Foster’s rule” (Foster 1964) and as “island syndrome” (Adler and Levins 1994). Island syndrome includes effects on several characteristics, including morphology and behavior, and there have been multiple explanations for why island populations differ from their mainland conspecifics, including reduced predator pressure and increased intraspecific competition on islands (Adler and Levins 1994). While island syndrome has been well studied with respect to demography, behavior, and morphology across several taxa (Novosolov et al. 2013; Benítez-López et al. 2021), fewer studies have explored the physiological implications of island life. Since ecological conditions and behavior are closely linked with endocrine-based responses, it has been hypothesized that island syndrome should also include changes to glucocorticoid levels, primarily associated with relatively low predation pressure on islands (Stewart et al. 2020). Relatively few studies have considered how glucocorticoid levels might differ between island and mainland conspecifics, although there is evidence for such shifts in birds (Clinchy et al. 2004; Müller et al. 2007). Given that glucocorticoids and tameness appear to be linked (i.e., with tameness dampening stress-induced endocrine responses; Rödl et al. 2007; Vitousek et al. 2010), differences in glucocorticoid levels between island and mainland populations could provide

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